

Telecom Churn Retention Decision Model

Model Card

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Course: BAN6440 Machine Learning

Repository: github.com/fruStoic/BAN6440_Final_Project

1. Model Overview

Field	Value
Model type	Logistic Regression
Task	Binary churn-risk estimation
Positive class	Customer churn
Calibration	Sigmoid calibration
Operational threshold	0.40 calibrated churn probability
Development protocol	5-fold stratified CV repeated 3 times
Primary selection metric	PR-AUC / Average Precision
Raw predictive inputs	17
Encoded features	28
Excluded predictive variables	gender, SeniorCitizen
Deployment mode	Weekly batch scoring

The model estimates customer churn probability and supports a retention-campaign decision. It is not intended as a general-purpose customer classification system. A customer is selected for intervention when the calibrated churn probability is at least 0.40 under the stated campaign assumptions.

2. Intended Use

- Prioritize customers for a scheduled telecommunications retention campaign.
- Support, not replace, human decisions about campaign policy and offer design.
- Use calibrated probabilities together with campaign economics rather than a fixed top- k list.

Out-of-scope uses: credit decisions, service denial or restriction, autonomous pricing or contract changes, demographic inference, or causal claims about campaign effectiveness.

3. Data and Preprocessing

The supplied dataset contains 7,043 customers, including 1,869 churners (26.54%). A fixed 80/20 stratified split produced 5,634 development observations and a 1,409-customer final holdout. The holdout was opened only after the pipeline was frozen.

- **customerID** is excluded from prediction.
- **TotalCharges** is converted from text to numeric; blank values are accepted only where **tenure** = 0, then assigned 0.
- Numeric variables are standardized with **StandardScaler**.
- Categorical variables use one-hot encoding with the first level dropped and unknown categories ignored.
- **gender** and **SeniorCitizen** are excluded from prediction but retained for fairness auditing.

4. Model Selection

Mean development PR-AUC values were 0.2654 for the majority baseline, 0.6549 for logistic regression, 0.6425 for HistGradientBoosting, and 0.6532 for the incumbent ANN. The predictive models were practically tied under the pre-registered paired-fold rule, so logistic regression was selected because additional complexity did not provide a meaningful predictive advantage.

5. Final Holdout Performance

Metric	Estimate	95% bootstrap CI
Accuracy	0.7899	0.7693 to 0.8119
Precision	0.5942	0.5459 to 0.6437
Recall	0.6578	0.6081 to 0.7062
F1	0.6244	0.5845 to 0.6650
PR-AUC	0.6484	0.5958 to 0.7019
ROC-AUC	0.8470	0.8255 to 0.8680
Brier score	0.1356	0.1260 to 0.1452

At the 0.40 threshold, the confusion matrix was TN=867, FP=168, FN=128, TP=246. The model selected 414 of 1,409 customers (29.38%).

6. Business Decision Rule

Base-case planning assumptions are: contact cost \$2, offer cost \$65, offer acceptance 40%, incremental retention uplift 15%, contribution margin 30% of a \$65 monthly revenue proxy, and a 24-month value horizon. These are planning assumptions, not dataset facts.

The exact break-even probability is 0.398860 and the pre-registered operational threshold is 0.40. On the final holdout, the 0.40 rule produced an expected campaign cost of \$11,592.00, expected benefit of \$17,073.27, expected net value of \$5,481.27, and expected ROI of 47.28%. These are expected values, not realized financial results.

7. Fairness and Segment Performance

Group	n	Prec.	Recall	FPR	FNR	Cal. gap
Female	722	58.99%	67.72%	16.70%	32.28%	+0.42 pp
Male	687	59.90%	63.78%	15.74%	36.22%	-0.74 pp
Non-senior	1,168	55.18%	62.03%	14.86%	37.97%	+1.12 pp
Senior	241	70.43%	75.00%	25.56%	25.00%	-6.30 pp

The SeniorCitizen group showed a higher false-positive rate and poorer calibration. These differences require monitoring but do not by themselves establish discriminatory model behavior. The protected variables are not predictive inputs.

8. Known Limitations

- The final holdout is a fresh internal split, not independent external validation; the same source dataset had been studied in Module 6.
- The dataset contains no timestamps, so temporal generalization cannot be tested.
- The 15% retention-uplift assumption is not estimated from the supplied data.
- The model predicts churn risk, not causal response to treatment.
- Expected ROI is sensitive to uplift and customer-value assumptions.
- Production drift may arise from changes in behavior, pricing, competitors, or campaign policy.

9. Production Monitoring and Governance

Monitor schema failures, missing values, feature drift, prediction drift, churn prevalence, Brier score, group-level calibration, precision/recall, FPR/FNR, campaign selection rate, batch completion, score freshness, model version, treatment status, campaign response, and observed churn outcome. Alerts should trigger investigation rather than automatic retraining.

The production artifact should version the feature policy, preprocessing, logistic model, calibrator, 0.40 threshold, and dependency versions together. Treatment assignment and campaign outcomes must be retained to reduce intervention bias and support future uplift analysis.